Can teleseismic data resolve a finite-fault model of an M_w 6.1?

The 2023 Mariana Islands earthquake (us6000kzv3) as a resolution test of the MTUQ+WISP pipeline

Automated pipeline (17.Venezuela_2026)

2026-07-29

Purpose

USGS publishes finite-fault models routinely for $M_\mathrm{w} \gtrsim 6.5$; for the 2023 M_w 6.1 Mariana Islands earthquake no finite-fault product exists. Hartzell, Mendoza & Zeng (2013, GRL) nevertheless resolved the M_w 5.8 Mineral, Virginia rupture from teleseismic P waves alone — their enabling ingredients being P waves to ∼1 Hz (attenuation $t^* \approx 1$ s still passes source detail at teleseismic range), resolution carried by timing rather than amplitude, km-scale subfaults, and a compact fault. This report applies that recipe with the MTUQ+WISP pipeline and states explicitly what is, and is not, resolved.

1 Event and reference solution

Origin	2023-08-14T13:51:53.8 UTC, 13.347°N, 147.530°E, depth 8 km
Magnitude	M_w 6.1 (W-phase); $M_0 = 1.74 \times 10^{18}$ N m; centroid 13.5 km; DC 91 %
Tectonics	oblique normal + strike-slip east of Guam, near the trench (outer-rise style)
W-phase NP1	253.3/56.1/−28.7
W-phase NP2	000.3/66.5/−142.6
USGS products	MT, ShakeMap, PAGER, DYFI — no finite fault

2 Data

205 three-component stations, 32–89°, maximum azimuthal gap 28° — the densest coverage of any event in this project, which matters doubly here: the mechanism has a large strike-slip component, and the Kumamoto blind test showed clustered azimuths alias strike-slip radiation. MTUQ used an azimuth-balanced 60-station response-corrected subset; WISP applied its own quality control. WISP's body-wave filter is 0.01–1.0 Hz — the Hartzell band — with auto-sized 2.4 × 2.0 km subfaults (255 total).

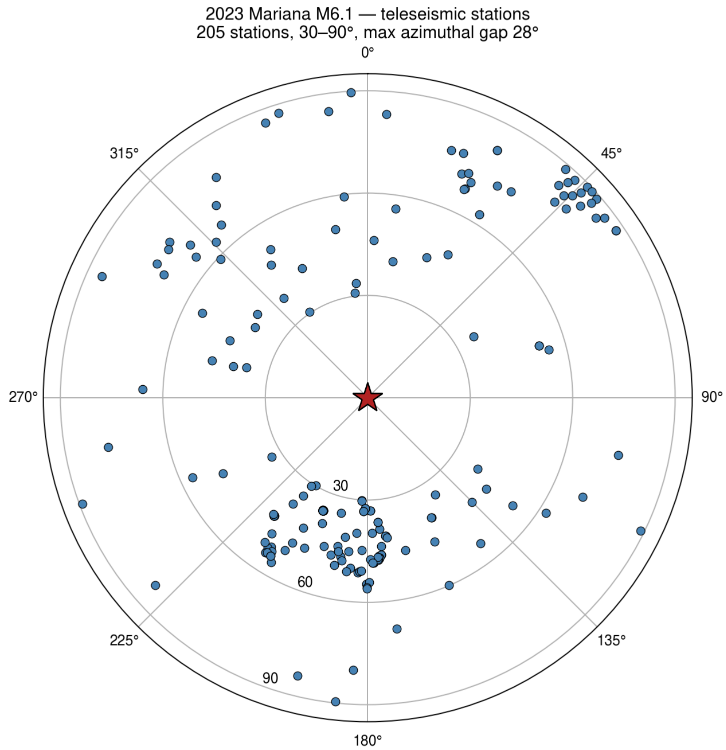


Figure 1: Teleseismic stations (azimuthal-equidistant; rings 30/60/90°).

3 Moment tensor (MTUQ): the independent point source fails at M_w 6.1

At $M_0 \sim 2 \times 10^{18}$ N m the standard teleseismic bands (body 10–33 s, surface 50–100 s) are low-SNR at 30–90°, and it shows:

variant	mechanism	M_w	Kagan vs W-phase
DC joint, 60 stations	68/82/−2	6.25	52.5°
DC body P+SH, 60 stations	122/54/52	5.80 (lower grid edge)	85.7°
depth scan 6–33 km	mechanism drifts	—	misfit varies 0.2 %

Table 1: Neither the mechanism nor the centroid depth is independently resolvable; the depth-misfit curve is flat to 0.2 % over 6–33 km. This is not an azimuth-coverage problem (gap 28°) — it is a signal-to-noise floor. Pipeline rule for $M_\mathrm{w} \lesssim 6.5$: take mechanism and depth from the W-phase and spend the teleseismic data on the finite fault — exactly the Hartzell strategy.

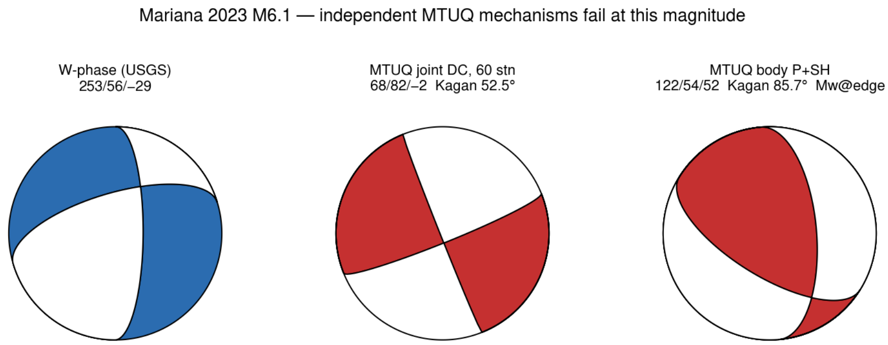
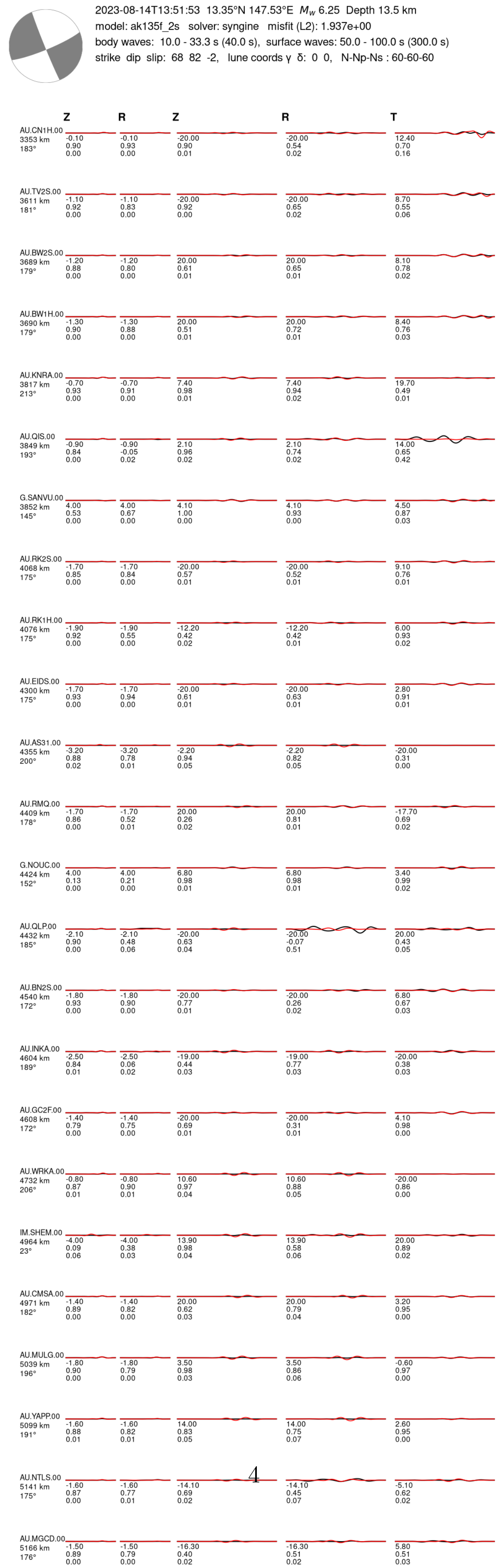


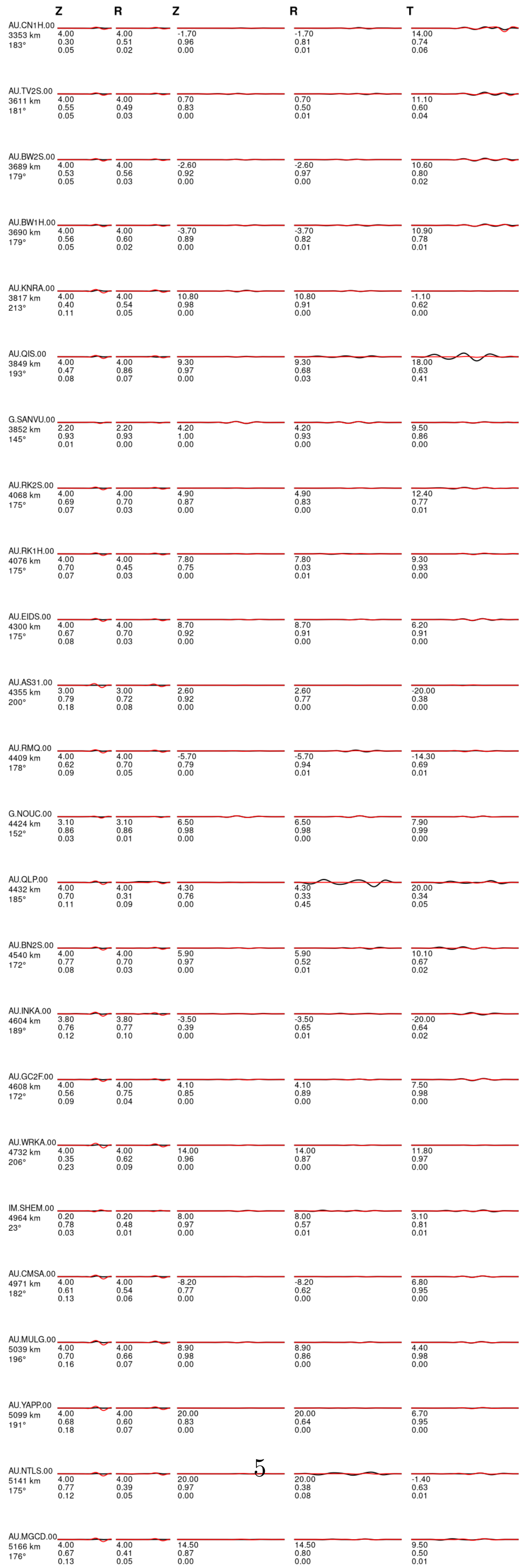
Figure 2: W-phase reference vs the two MTUQ attempts.

Why the MTUQ result is poor: the waveforms show it directly

The fits below make the failure mode concrete. Observed peak amplitudes are 1–5 nm (body) and a few mm (long-period surface waves) — at or barely above the noise floor of the teleseismic bands. Most of the ∼180 traces are near-flat; a handful carry large oscillations that are noise bursts, not signal, and the normalized L2 misfit is then steered by those few traces (the low-effective-trace-count failure familiar from the regional Korea pipeline). Crucially, evaluating the W-phase mechanism itself on the same processed data (second figure) fits the usable traces essentially as well as the grid-search best — the misfit surface is flat between mechanisms 50° apart, i.e. the data do not discriminate. The poor MTUQ result is a property of the magnitude, not of the protocol.


2023-08-14T13:51:53 13.35°N 147.53°E M_w 6.25 Depth 13.5 km
model: ak135f_2s solver: syngine misfit (L2): 1.937e+00
body waves: 10.0 - 33.3 s (40.0 s), surface waves: 50.0 - 100.0 s (300.0 s)
strike dip slip: 68 82 -2, lune coords γ δ: 0 0, N-Np-Ns : 60-60-60
Z R Z R T
AU.CN1H.00 3353 km 183° -0.10 0.90 0.00 -0.10 0.93 0.00 -20.00 0.90 0.01 -20.00 0.54 0.02 12.40 0.70 0.16
AU.TV2S.00 3611 km 181° -1.10 0.92 0.00 -1.10 0.83 0.00 -20.00 0.92 0.00 -20.00 0.65 0.02 8.70 0.55 0.06
AU.BW2S.00 3689 km 179° -1.20 0.88 0.00 -1.20 0.80 0.00 20.00 0.61 0.01 20.00 0.65 0.01 8.10 0.78 0.02
AU.BW1H.00 3690 km 179° -1.30 0.90 0.00 -1.30 0.88 0.00 20.00 0.51 0.01 20.00 0.72 0.01 8.40 0.76 0.03
AU.KNRA.00 3817 km 213° -0.70 0.93 0.00 -0.70 0.91 0.00 7.40 0.98 0.01 7.40 0.94 0.02 19.70 0.49 0.01
AU.QIS.00 3849 km 193° -0.90 0.84 0.00 -0.90 -0.05 0.02 2.10 0.96 0.02 2.10 0.74 0.02 14.00 0.65 0.42
G.SANVU.00 3852 km 145° 4.00 0.53 0.00 4.00 0.67 0.00 4.10 1.00 0.00 4.10 0.93 0.00 4.50 0.87 0.03
AU.RK2S.00 4068 km 175° -1.70 0.85 0.00 -1.70 0.84 0.00 -20.00 0.57 0.01 -20.00 0.52 0.01 9.10 0.76 0.01
AU.RK1H.00 4076 km 175° -1.90 0.92 0.00 -1.90 0.55 0.00 -12.20 0.42 0.02 -12.20 0.42 0.01 6.00 0.93 0.02
AU.EIDS.00 4300 km 175° -1.70 0.93 0.00 -1.70 0.94 0.00 -20.00 0.61 0.01 -20.00 0.63 0.01 2.80 0.91 0.01
AU.AS31.00 4355 km 200° -3.20 0.88 0.02 -3.20 0.78 0.01 -2.20 0.94 0.05 -2.20 0.82 0.05 -20.00 0.31 0.00
AU.RMQ.00 4409 km 178° -1.70 0.86 0.00 -1.70 0.52 0.01 20.00 0.26 0.02 20.00 0.81 0.01 -17.70 0.69 0.02
G.NOUC.00 4424 km 152° 4.00 0.13 0.00 4.00 0.21 0.00 6.80 0.98 0.01 6.80 0.98 0.01 3.40 0.99 0.02
AU.QLP.00 4432 km 185° -2.10 0.90 0.00 -2.10 0.48 0.06 -20.00 0.63 0.04 -20.00 -0.07 0.51 20.00 0.43 0.05
AU.BN2S.00 4540 km 172° -1.80 0.93 0.00 -1.80 0.90 0.00 -20.00 0.77 0.01 -20.00 0.26 0.02 6.80 0.67 0.03
AU.INKA.00 4604 km 189° -2.50 0.84 0.01 -2.50 0.06 0.02 -19.00 0.44 0.03 -19.00 0.77 0.03 -20.00 0.38 0.03
AU.GC2F.00 4608 km 172° -1.40 0.79 0.00 -1.40 0.75 0.00 -20.00 0.69 0.01 -20.00 0.31 0.01 4.10 0.98 0.00
AU.WRKA.00 4732 km 206° -0.80 0.87 0.01 -0.80 0.90 0.01 10.60 0.97 0.04 10.60 0.88 0.05 -20.00 0.86 0.00
IM.SHEM.00 4964 km 23° -4.00 0.09 0.06 -4.00 0.38 0.03 13.90 0.98 0.04 13.90 0.58 0.06 20.00 0.89 0.02
AU.CMSA.00 4971 km 182° -1.40 0.89 0.00 -1.40 0.82 0.00 20.00 0.62 0.03 20.00 0.79 0.04 3.20 0.95 0.00
AU.MULG.00 5039 km 196° -1.80 0.90 0.00 -1.80 0.79 0.00 3.50 0.98 0.03 3.50 0.86 0.06 -0.60 0.97 0.00
AU.YAPP.00 5099 km 191° -1.60 0.88 0.01 -1.60 0.82 0.01 14.00 0.83 0.05 14.00 0.75 0.07 2.60 0.95 0.00
AU.NTLS.00 5141 km 175° -1.60 0.87 0.00 -1.60 0.77 0.01 -14.10 0.69 0.02 -14.10 0.45 0.07 -5.10 0.62 0.02
AU.MGCD.00 5166 km 176° -1.50 0.89 0.00 -1.50 0.79 0.00 -16.30 0.40 0.02 -16.30 0.51 0.02 5.80 0.51 0.03

2023-08-14T13:51:53 13.35°N 147.53°E M_w 6.09 Depth 13.5 km
model: ak135f_2s   solver: syngine   misfit (L2): 2.047e+00
body waves: 10.0 - 33.3 s (40.0 s), surface waves: 50.0 - 100.0 s (300.0 s)
strike dip slip: 253 56 -29,   lune coords γ δ: 0 0,   N-Np-Ns : 60-60-60



4 Finite fault (WISP): both W-phase planes, Mww-seeded, **`shift_match`** applied

plane	strike/dip/rake	misfit (post-shift)	peak slip	T_{5-95}
NP1 (preferred)	253/56/−29	0.2012	24.9 cm	8.2 s
NP2	0/66/−143	0.2058	24.9 cm	7.9 s

Table 2: Recovered $M_0 = 1.76 \times 10^{18}$ N m vs W-phase 1.74×10^{18} — 1 % agreement. The 2.2 % NP1 preference sits between the Calama near-tie (1.6 %, ∼10 km rupture) and the SPdA decisive margin (10 %, ∼50 km rupture): suggestive, not conclusive, exactly as the rupture-size criterion predicts.

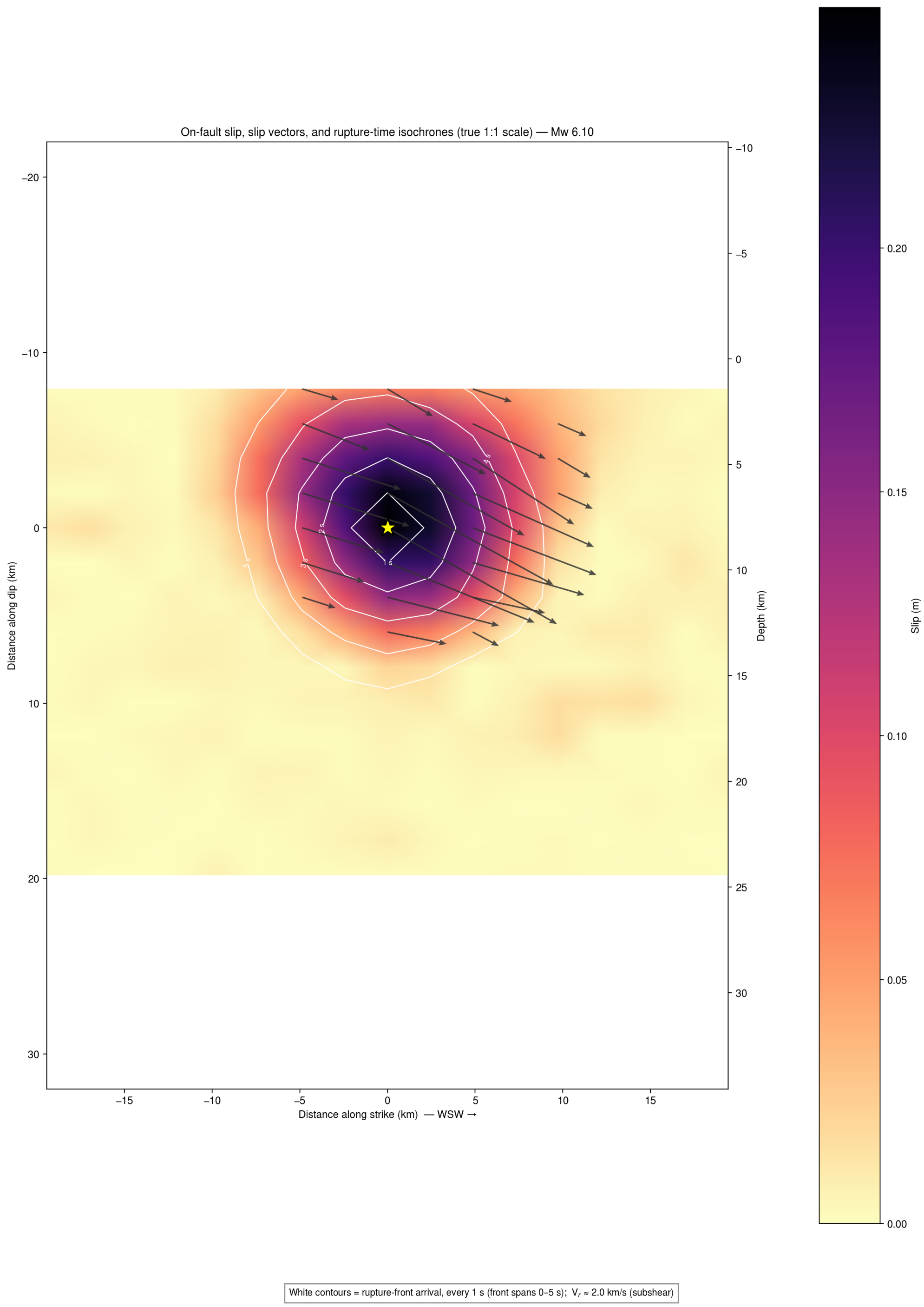


Figure 5: Preferred plane (253/56/−29), true 1:1 scale: a single compact asperity ∼10 km across at 2–13 km depth, centred on the 8-km hypocentre; peak slip 25 cm; concentric 1-s isochrones spanning 0–5 s ($V_r \approx 2.0$ km s^{-1}, subshear); oblique right-lateral+normal slip vectors.

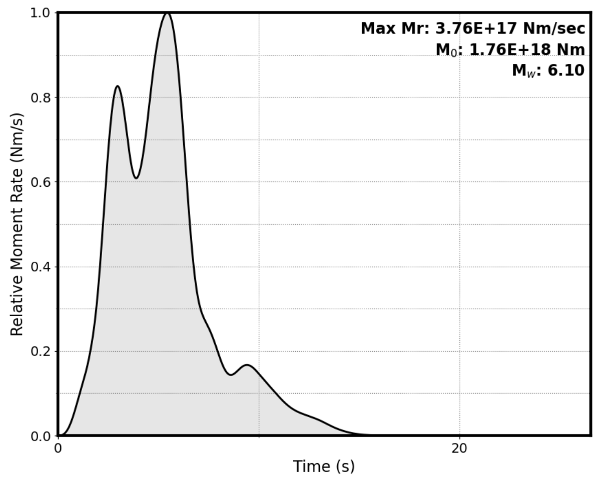


Figure 6: Moment-rate function (preferred plane).

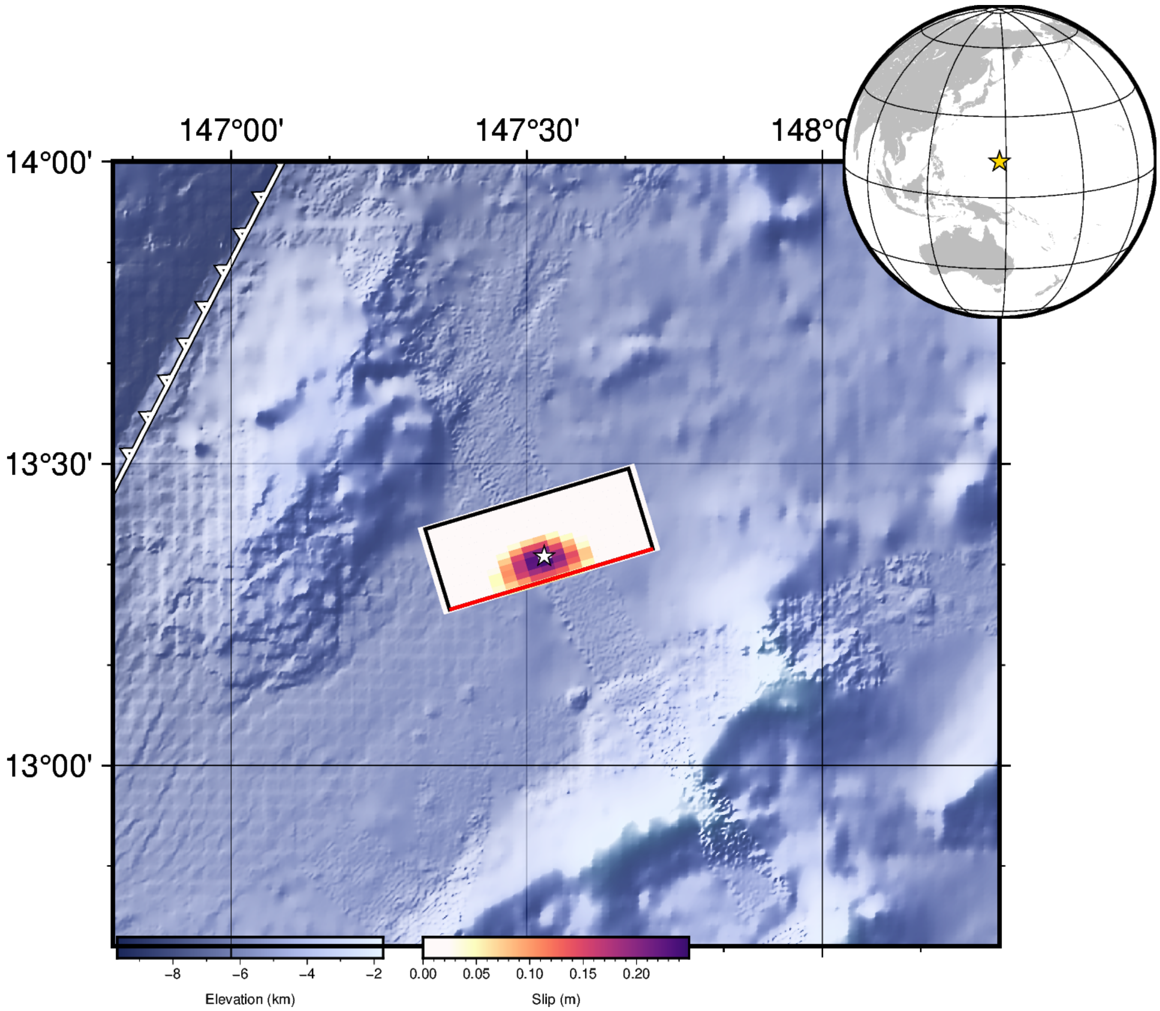


Figure 7: Map view.

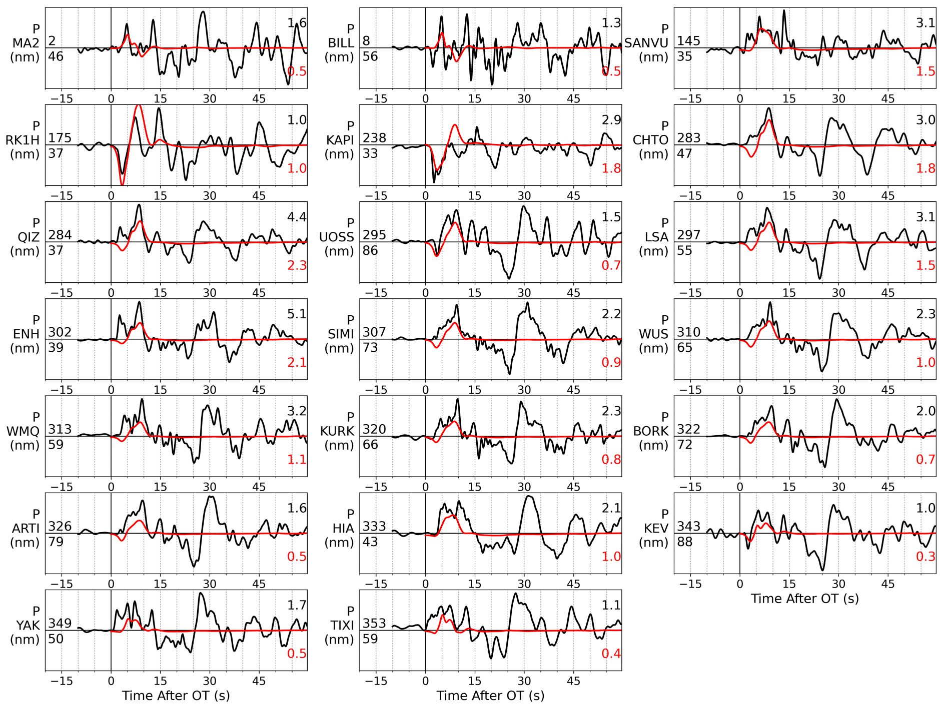


Figure 8: Teleseismic P fits (preferred plane, shifted) — the dataset that carries the Hartzell-style timing resolution.

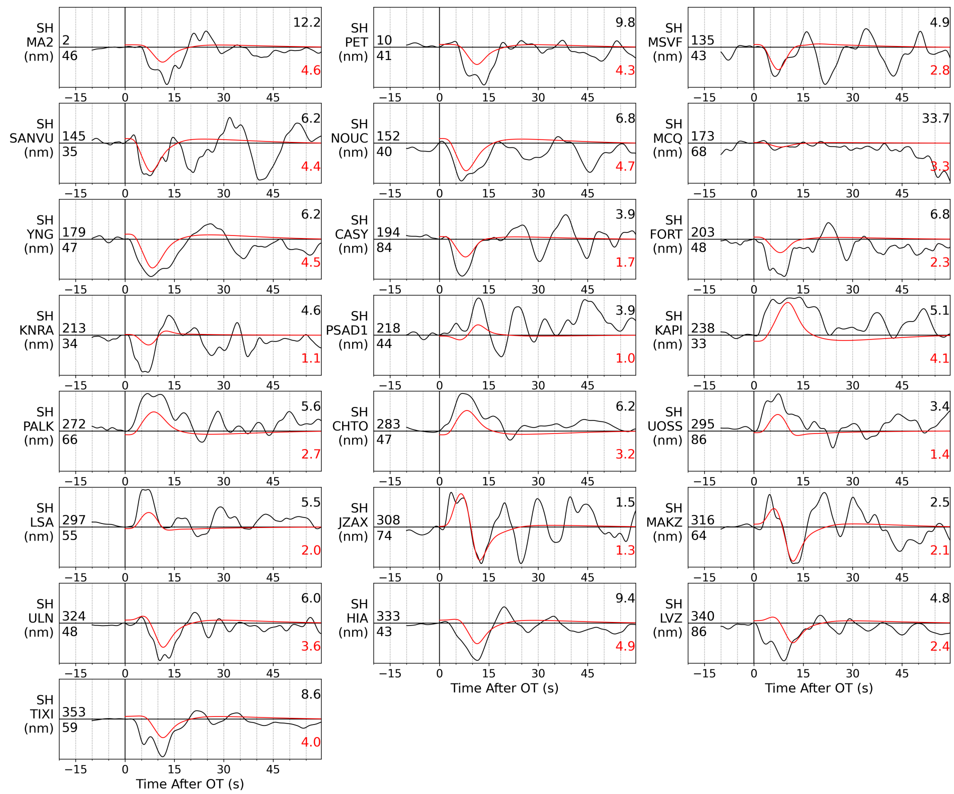


Figure 9: Teleseismic SH fits (preferred plane, shifted).

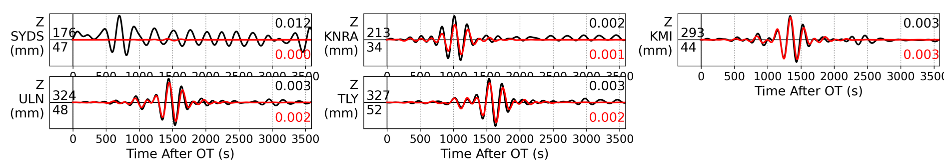


Figure 10: Rayleigh fits. WISP's quality control retained only five usable long-period vertical channels — mm-scale mantle waves sit at the noise floor for M_w 6.1 — but the retained traces fit well (SYDS is a zero-weight rejected trace shown for completeness). Surface waves constrain moment, not slip detail, at this magnitude.

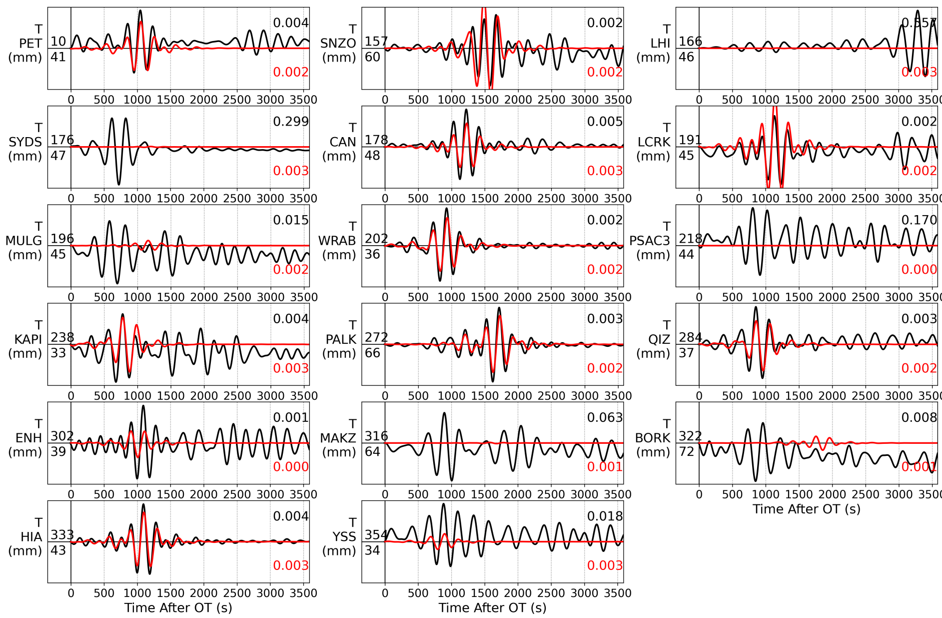


Figure 11: Love fits (same caveat).

Why the body-wave window after ∼15 s is rarely fit

A systematic feature of the P fits: the direct pulse (0–10 s, including pP/sP at 4.5–6.5 s for the 13.5-km source) is matched at nearly every station, while observed energy after ∼15 s is left unfit. Two causes, both identifiable:

1. The velocity model has no ocean. The source-side model (table below) tops out in a 90-m soft-sediment layer — but this trench-adjacent source lies beneath ∼4–5 km of Pacific water. Water-column multiples (pwP and its reverberations, two-way time $2 \times 4.5/1.5 \approx 6$ s per bounce) form trains persisting tens of seconds that a land-style 1-D model cannot produce at all. This is a known limitation of the operational WISP crust model for oceanic events, not a wrong crustal layer.

2. The oceanic microseism band. Several stations show the same 5–8-s oscillations before the P arrival at the same nm amplitude — the double-frequency microseism peak lives at exactly these periods and amplitudes, so part of the late window is simply noise that no source model should fit.

V_p (km/s)	V_s (km/s)	ρ (g/cm^3)	thickness (km)	
2.56	0.60	1.82	0.09	(soft sediment — no water layer above)
5.37	2.97	2.42	1.12	
6.45	3.67	2.71	2.45	
7.36	4.20	2.90	7.57	
8.08	4.47	3.38	196	

Table 3: WISP source-side velocity model. The absence of a water column is the structural reason the post-15-s window cannot be fit.

The practical consequence is benign for the headline conclusions — the inversion is anchored by the direct-pulse timing (the Hartzell resolution carrier), and the plane-stable STF confirms it — but it inflates the absolute misfit (0.20 vs 0.15 for continental Kumamoto) and is the first thing to fix (add a water layer / use an oceanic crust profile) if slip detail beyond first order is ever needed for a trench event.

5 Resolution assessment

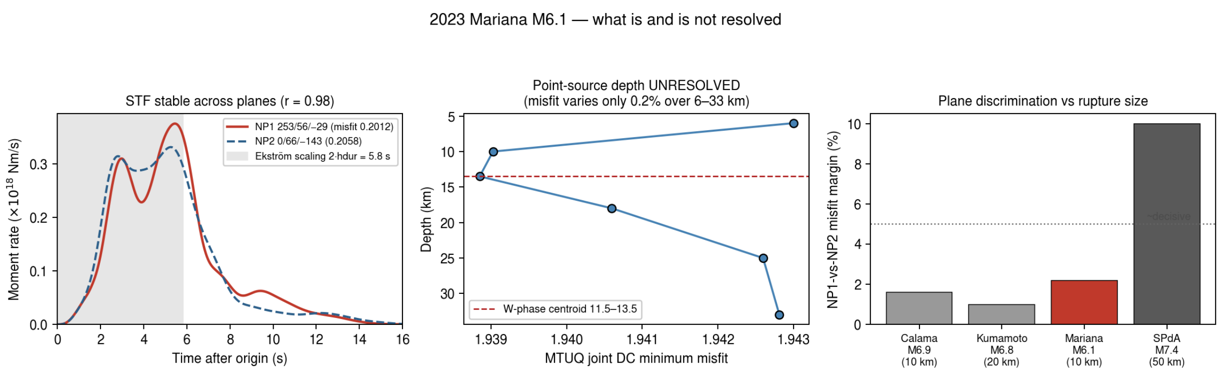


Figure 12: Left: the source-time function is stable across the two (very different) assumed planes, $r = 0.98$, including its two-pulse structure at 2.8 and 5.4 s — the plane-independent, therefore data-driven, part of the model. The Ekström-scaling duration band ($2 \cdot h_{dur} = 5.8$ s) captures the STF body; the tail beyond it is regularization. Centre: the MTUQ point-source depth scan is flat to 0.2 % — depth must come from the W-phase. Right: the NP1-vs-NP2 misfit margin in the context of the pipeline's plane-discrimination criterion.

Resolved (stable across planes, passing physical checks):

- a single compact asperity ∼10 km across at 2–13 km depth, centred on the hypocentre — consistent with M_w 6.1 source scaling;
- the STF shape including its two-pulse structure (2.8 + 5.4 s), cross-plane $r = 0.98$;
- M_0 to 1 % of the W-phase; rupture front spanning 0–5 s, $\mathrm{V}_r \approx 2.0$ km s^{-1};
- stress drop 1.0 MPa (effective-area convention, $(\Sigma u)^2/\Sigma u^2 \to 263$ km^2) to 3.7 MPa (half-peak area, 110 km^2) — in the physically expected 1–10 MPa range.

Not resolved (and stated so):

- the fault plane: the 2.2 % NP1 margin is suggestive, not decisive, at 10-km rupture size;
- an independent point-source mechanism or centroid depth (Section 3);
- fine slip detail and the duration tail: T_{5-95} = 8.2 s exceeds the 5.8-s scaling expectation because smoothing spreads low-level slip; the 25-cm peak is a lower bound;
- absolute updip/downdip placement within the compact patch.

6 Verdict

Yes: a reliable first-order finite-fault model of an M_w 6.1 is recoverable from teleseismic data — when the Hartzell recipe is followed (W-phase-fixed mechanism, P waves to 1 Hz, km-scale subfaults, claims restricted to plane-stable features). What cannot be claimed at this magnitude: the fault plane, an independent mechanism, or a centroid depth from teleseismic point-source inversion. This extends the pipeline's validated range from M_w 6.8 down to M_w 6.1, one magnitude unit above the Hartzell et al. (2013) M_w 5.8 benchmark.